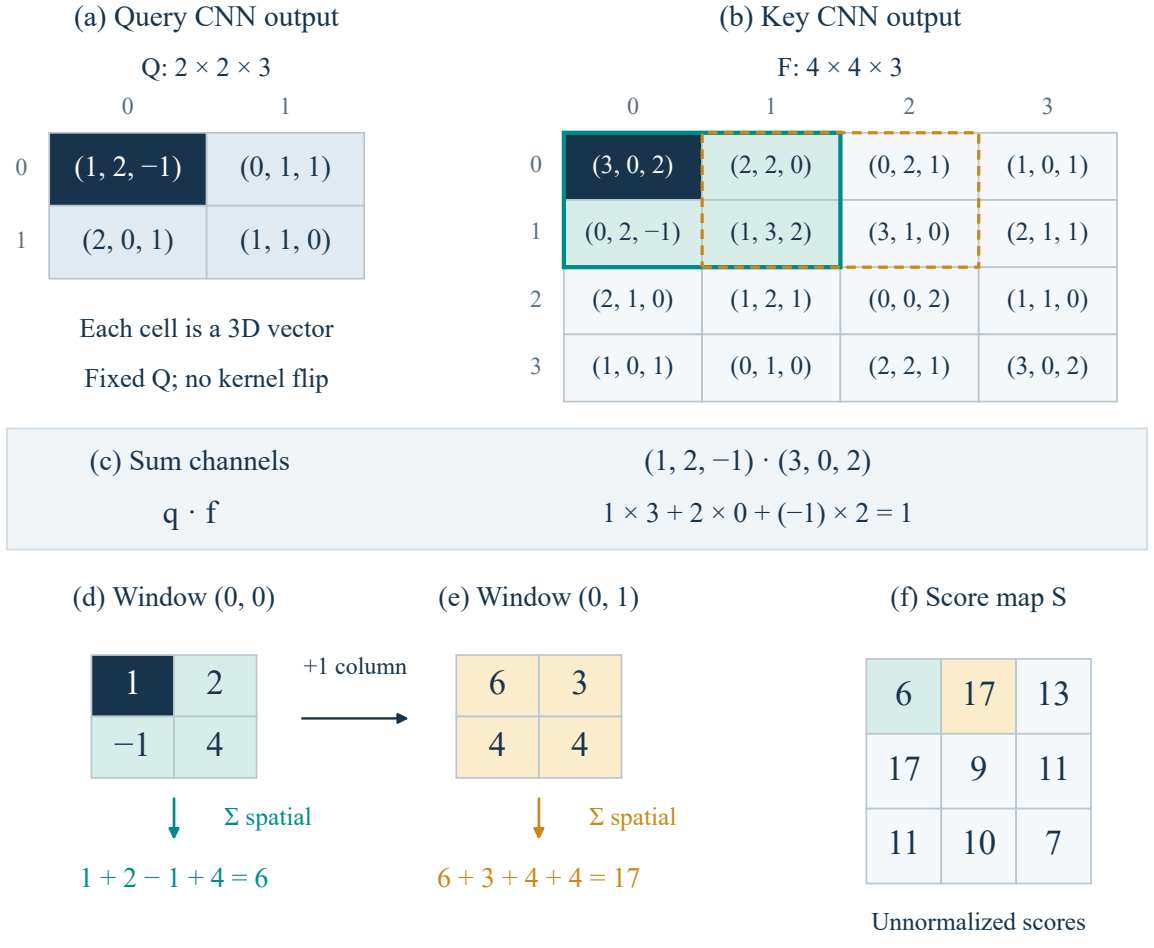
(a) Query CNN output
Q: 2 × 2 × 3
0
1
0
1
(1, 2, −1)
(0, 1, 1)
(2, 0, 1)
(1, 1, 0)
Each cell is a 3D vector
Fixed Q; no kernel flip
(b) Key CNN output
F: 4 × 4 × 3
0
1
2
3
0
1
2
3
(3, 0, 2)
(2, 2, 0)
(0, 2, 1)
(1, 0, 1)
(0, 2, −1)
(1, 3, 2)
(3, 1, 0)
(2, 1, 1)
(2, 1, 0)
(1, 2, 1)
(0, 0, 2)
(1, 1, 0)
(1, 0, 1)
(0, 1, 0)
(2, 2, 1)
(3, 0, 2)
(c) Sum channels
q · f
(1, 2, −1) · (3, 0, 2)
1 × 3 + 2 × 0 + (−1) × 2 = 1
(d) Window (0, 0)
1
2
−1
4
+1 column
Σ spatial
1 + 2 − 1 + 4 = 6
(e) Window (0, 1)
6
3
4
4
Σ spatial
6 + 3 + 4 + 4 = 17
(f) Score map S
6
17
13
17
9
11
11
10
7
Unnormalized scores
Illustrative features, not model activations · Single orientation · Stride 1, no padding